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Influence of Maternal Proximity on Behavioral and Physiological Responses to Separation in Infant Rhesus Monkeys (*Macaca mulatta*)

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The effects of maternal proximity on the behavioral and physiological responses of infant rhesus macaques during 4 days of total or adjacent separations from the mother were studied. The 6 infants tested showed behavioral responses that differentiated the two separation conditions. Major differences were found in the quantity and quality of vocalizations, the occurrence of cage-biting and cage-shaking behavior, object exploration, and hunched and freezing postures. In particular, the structure of coo vocalizations clearly discriminated between the presence or the absence of the mother during separation. Cerebrospinal fluid (CSF) concentrations of dopamine and serotonin metabolites did not discriminate between the two separation conditions but showed a transient elevation at 24 hr after separation and were not different from baseline by 96 hr after separation. In contrast, both the plasma cortisol and the CSF norepinephrine metabolite responses tended to be greater and to persist for a longer period of time when infants were totally isolated. The results are discussed within the context of attachment and coping theories.

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The consequences of maternal separation and loss during infancy have been of major interest in primate research during the past 30 years. One area of concern has been the study of the infant's response to separation and its potential link to depression and other mental illness (Kraemer, Ebert, & McKinney, 1983; Mineka & Suomi, 1978). Another focus of the separation model has been the study of the psychobiology of attachment in the mother–infant dyad (the response of physiological systems to this psychological manipulation). For example, it has been shown that, in the case of brief maternal separation, specific parameters of separation can greatly influence the behavioral as well as the endocrine responses to separation in infant rhesus or squirrel monkeys. These studies examined the influence of social variables, including maternal proximity, on standardized measures of behavior and hormones in infants exposed to different conditions of separation (Levine & Wiener, 1988).

In one experiment (Levine, Johnson, & Gonzalez, 1985), the response of rhesus infants separated totally from their mothers (total isolation) was compared with the response of rhesus infants separated physically (but in visual, auditory, and olfactory contact) from their mothers (adjacent separation). The results indicated that although the infants separated totally from their mothers called less, as measured by the rate of vocalizations emitted after separation, they showed a greater level of arousal, as measured by plasma cortisol levels, than infants separated adjacent to their mothers. It was suggested that increased calling behavior and an attenuated cortisol response like that observed during the adjacent separation may have adaptive value for the infant. This finding was interpreted within the context of the coping theory that predicts that under stressful conditions, proximity-seeking behavior may result in the modulation and reduction of the infants' arousal levels (Levine, 1983). In this case, calling to adjacent mothers is the proximity-seeking behavior that lowers cortisol levels, a measure of the infants' arousal. This finding also led to a reevaluation of the concept of the "protest phase" that was originally developed to describe the infant distress response to maternal separation (Levine, Franklin, & Gonzalez, 1984).

In this study we examined the effects of maternal proximity on the behavioral and physiological responses of infant rhesus macaques (*Macaca mulatta*) to maternal separation using a separation paradigm similar to the one described previously (adjacent vs. total separation), but for a longer period of time (96-hr duration). The purpose of this study was first to replicate the experiment that revealed an apparent inverse relationship between the infant's calling behavior and its state of arousal following these two separation conditions. Second, we were interested in examining more closely the behavioral responses, particularly vocalizations, of the infants after sep-

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aration from their mothers. This study involved a systematic analysis of the quantity as well as the quality of the calls emitted by the infants in the two separation conditions. We were also interested in behavioral changes occurring during the 4 days of separation in each of the two conditions. Finally, we were particularly interested in studying changes in brain monoamines activity (as measured by their metabolites in cervical cerebrospinal fluid) that could be associated with behavioral changes, changes in pituitary–adrenal activity, or both, following these two separation conditions. Several hypotheses regarding the function of brain neurotransmitters indicate that monoamines are very often associated with the motor, affective, and cognitive processes of the organism (Carlsson, 1987). Changes in cerebrospinal fluid (CSF) monoamine metabolite concentrations reflect changes in brain neurotransmitter activity. Thus, the comparison of monoamine metabolite concentrations in the infant's CSF before and after separation, as well as between separation conditions, could be of great value for assessing the central state of the infant after maternal separation, as it correlates to behavior.

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Method

Subjects

The 6 mother–infant dyads (*M. mulatta*) used in this study came from two social groups (that had been established at Stanford University for approximately 10 years). The groups were maintained in two adjacent outdoor cages (5 × 10 × 3 m) separated by a chain-link fence that adjoined smaller indoor quarters (3 × 5 × 3 m). One group consisted of 1 adult male, 11 adult females, and 5 infants (4 male, 1 female). The other group consisted of 1 young adult male, 8 adult females, and 1 infant female. The 6 infants subjected to separation were between 6 and 8 months of age during the study.

After removal from their social groups and during separation, the mothers and infants were housed in standardized cages (0.75 × 0.75 × 1.25 m) in temperature-controlled (21 ± 2 °C) observation rooms on a natural light schedule. The experiment was conducted between November and February. Food (Wayne 25% Monkey Diet, Allied Mills, Inc., Chicago, IL) and water were available ad lib, and fruit was provided daily for all separated infants and mothers.

Procedure

Each of the subjects was studied under two separation conditions: either in total isolation (infant removed from the mother and placed alone in a cage in another room) or in adjacent separation (infant removed from the mother and placed in a cage adjacent to the mother, allowing visual and auditory access but not tactile contact). Each separation was 96 hr in duration, and the period between separations was 10 days.

Two dyads at a time were captured from the social groups and placed in two different rooms where they stayed for 1 week (adaptation period), at the end of which the mothers and infants were anesthetized with ketamine, and blood and CSF samples were taken from the infants (Basal Sample 1). Three days later, the infants were separated from their mothers at between 10:00 a.m. and 10:30 a.m. and placed into one of the two separation conditions (first separation period). Each infant was observed for 30 min immediately after separation and at the same time on each of the next 4 days. Two additional blood and CSF samples were taken from the anesthetized infants approximately 90 min after behavioral observations of the 2 infants at 24 and 96 hr of separation (stress samples). After the separation and samples, each infant was returned to its mother, and the dyads were switched to the opposite rooms. They were then allowed to stay 1 week undisturbed, at the end of which blood and CSF samples were again taken from the infants (Basal Sample 2). Three days later, each infant was subjected to the other separation condition for 96 hr again using the same procedure just described. At the end of the second separation period, mother and infant were reunited, and the dyads were returned to their social groups. Overall, the 6 infants were separated and observed under both conditions in a counterbalanced order.

Behavioral Recording

The behavioral measures were scored on a check sheet by an observer seated behind a one-way viewing window in front of the subject. Categories of behavior, reflective of the state of the animal, were scored in 30-s intervals either for frequency (vocalizations: coo calls, originally referred to as clear calls [Rowell & Hinde, 1962] or whoo calls [Hinde & Spencer-Booth, 1967], and screams, also referred to as screeches [Rowell & Hinde, 1962]; level of activity: number of times the animal crossed the midline of the cage) or for a single occurrence (object exploration, cage biting or cage shaking, eating, drinking, stereotypy, hunched-over and freezing postures, scratching, self-biting, and self-grooming). In addition, the calls emitted by the infants during separation were recorded during each behavioral observation period. The equipment used for recording consisted of either a portable cassette recorder (Sony TC110A) with an electret condenser microphone (Sony F25S cardioid; 4 infants) or a reel-to-reel tape recorder (Uher 4000 Report L at 9.5 cm/s) with an electret condenser microphone (Sony ECM-2300, shotgun with a variable directivity zoom; 2 infants). The distance between the microphone and the subject (approximately 1.5 m), as well as the level of recording or the zoom adjustment, were kept constant during all recording.

Physiological Measurements

The blood samples (0.5 ml) were obtained by saphenous venipuncture using heparinized syringes after immobilizing the infant with ketamine (25 mg). Immediately after the blood sample, CSF (0.2 ml) was collected by cervical spinal tap. The infant's neck was shaved and then sterilized with betadine solution and merthiolate. A 1-in (2.54 cm) 23-gauge standard injection needle with 1-ml syringe was carefully inserted perpendicular to the spine between the first two cervical vertebrae. After collection of CSF, the needle was removed, and the neck area was sprayed with merthiolate. The sampling procedure was carried out within 3 min of capture. Blood samples were centrifuged, and the plasma was collected and frozen at −20 °C until radioimmunoassay. CSF samples were stored at −80 °C until they were assayed for monoamine metabolite concentrations.

Cortisol Assay Procedures

Plasma cortisol was assayed using antiserum F21-53 (Endocrine Systems, Tarzana, CA) according to the method of Klemm and Gupta (1975). The use of this method in our laboratory has been previously described in detail by Smotherman, Hunt, McGinnis, and Levine (1979).

Monoamine Metabolite Assay Procedures

Measurements of monoamine metabolite concentrations were made using the procedure described by Faull, Anderson, Barchas, and

Berger (1979) but modified to accommodate the small sample sizes. CSF samples of 60 μl were analyzed in triplicate. An aqueous solution of internal standards (50 μl) containing 2-[3-(d_3-methoxy)-4-hydroxyphenyl]-1,2-dihydroxyethane ($^2H_3$-MHPG, 40 pmol, synthesized according to Markey, Powers, Dubinsky, & Kopin, 1979), 2-[3-(d_3-methoxy)-4-hydroxyphenyl]-acetic acid ($^2H_3$-HVA, 200 pmol), and 5-hydroxyindole-3-[2-d_2-acetic acid] ($^2H_2$-5-HIAA, 200 pmol) was added to each sample. Ascorbic acid was then added (100 μl, 14 mM), and the solutions were acidified (6 N HCl, 20 μl) and extracted with 1-ml volumes of freshly redistilled ethylacetate. The organic phases were pooled and evaporated to dryness in a stream of nitrogen. The dried residue was treated with pentafluoropropionic anhydride (40 μl) and pentafluoropropanol (10 μl, at 75 °C, 5 min). The residual reagent was again removed in a stream of nitrogen. The dried samples were stored at −80 °C and redissolved in 10 μl of ethylacetate prior to gas chromatography/mass spectrometry analysis (dropping needle injector, medium polarity bonded-phase fused-silica capillary column, electron impact ionization, HP 5985B quadrupole mass spectrometer).

Quantitative data were collected by recording the relative intensities of the pairs of ions at m/z 622/625, 460/463, and 615/617 corresponding to the molecular ions of the derivatives of unlabeled and labeled MHPG, HVA, and 5-HIAA, respectively. Calculation of the quantity of metabolite in each sample was made by interpolation from standard calibration curves prepared at the same time with the CSF samples. The calibration curves covered the typical concentration range for each metabolite (MHPG, 0–35 pmol; HVA, 0–137 pmol; 5-HIAA, 0-191 pmol) and showed excellent linearity with intercepts on the ordinate of less than 10%.

Statistical Analysis

Each behavior was examined independently and analyzed for differences between the two separation conditions across the 5 days of testing using a 2 (conditions) × 5 (times) analysis of variance (ANOVA) with repeated measures for time and condition. The Newman-Keuls post hoc test was used when appropriate ($p < .05$).

Preliminary analysis indicated three general categories of coo calls. Two were found in the adjacent condition (one call modulated in frequency [AM] and one call not modulated in frequency [A]) and one in the total isolation condition (T). The AM call differs from the A call by showing at least a 100-Hz frequency modulation over time in the fundamental band. For each subject, 12 vocalizations in each category were chosen at random during the first 30 min of recording after separation. These vocalizations were then subjected to spectrographic analysis using a Kay sonagraph (Model 6061B) with a narrow-band (45 Hz) frequency analyzer and a frequency range of 80–8000 Hz. In addition, amplitude changes were recorded using the amplitude display of the sonagraph.

Ten physical parameters (see definitions in Table 1) were measured directly on the sonagrams using a ruler with convertible scales for durations (±5 ms) and frequencies (±50 Hz). We then calculated the means plus standard deviations of each parameter for each type of call and for each subject tested.

Each parameter was looked at independently for differences between the scores under different conditions (AM, A, and T) or for differences between the scores of each subject. We used a one-way 6 (subjects) × 3 (conditions) ANOVA with repeated measures for conditions. Subsequent post hoc tests were done on cases with significant differences ($p < .05$).

Mean plasma cortisol levels and mean CSF monoamine metabolite concentrations from the 6 infants were subjected to analysis for differences between conditions (adjacent and total separation) and differences among base samples, 24-hr separation samples, and 96-hr separation samples using a two-way 2 (conditions) × 3 (times) ANOVA with repeated measures for time and condition. A Newman-Keuls post hoc test was used when necessary at a significance level of $p < .05$. In addition, the cortisol data were further analyzed, after conversion to differences from base values, using a two-way 2 (conditions) × 2 (times) ANOVA with repeated measures for time and condition.

Results

Behavioral Analysis

The infants continued to eat and drink throughout the separations, and their activity remained stable during the 4 days of separation in both conditions. The analysis of vo-

Table 1
Definitions of Acoustic Measures Used in the Analysis of Vocalizations

Term	Abbreviation	Definition
Duration	D	total duration of the call in milliseconds
Duration to peak	DP	duration of frequency modulation (from the beginning of the call to peak frequency) in milliseconds
Start frequency	SF	absolute value of fundamental at the beginning of the call in hertz
Peak frequency	PF	peak value of fundamental in hertz
End frequency	EF	absolute value of fundamental at the end of the call in hertz
Number of harmonics	NH	total number of harmonics in call
Formant	F	value of the most intense harmonic in hertz
Durations of amplitude modulation[a]	D1	duration of the first amplitude modulation (from the beginning of the call to the first point of inflection after a 50% decrease in amplitude after the first peak) in milliseconds
	D2	duration of the first two amplitude modulations (from the beginning of the call to the first point of inflection after a 50% decrease in amplitude after the second peak) in milliseconds
	D3	duration of the first three amplitude modulations (from the beginning of the call to the first point of inflection after a 50% decrease in amplitude after the third peak) in milliseconds

[a]Sonagrams of the amplitude modulations for examples of coo vocalizations are depicted in Figure 3.

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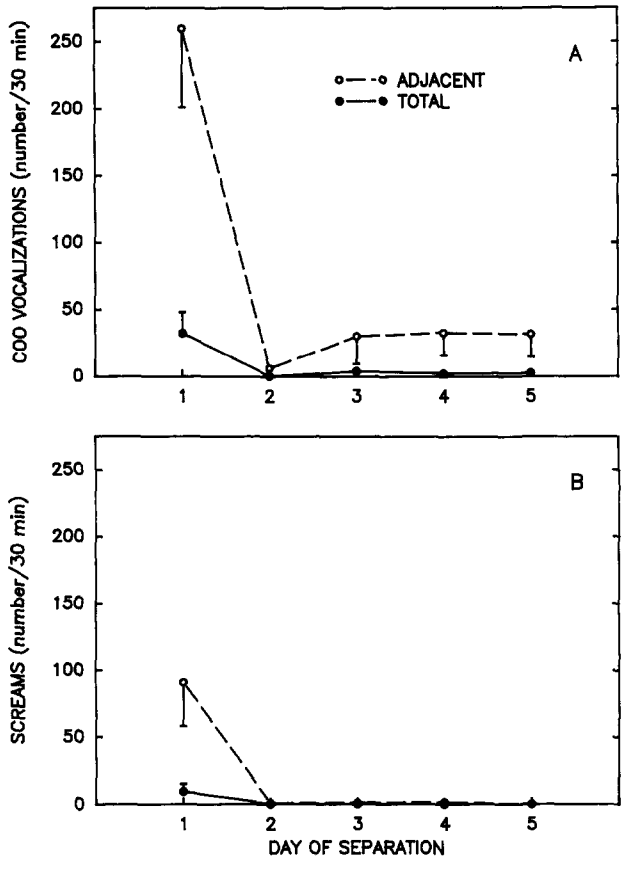


Figure 1. Mean (± *SE*) rates of vocalizations: (A) coo vocalizations and (B) screams emitted by the infants during adjacent and total separations.

calization rates shows a Condition × Time interaction for coo vocalizations, $F(4, 20) = 15.06$, $p < .001$, and screams, $F(4, 20) = 7.52$, $p < .001$. On the 1st day of separation, the vocalization rates in the adjacent and total conditions were significantly different. A significant decline in calling behavior was observed after the 1st day of separation in the adjacent condition. In contrast, the infants separated in the total separation condition emitted very few vocalizations throughout the whole separation period. Figure 1 shows the amount of coo vocalizations and screams emitted by the infants under the two separation conditions. Only coo vocalizations were subjected to qualitative analysis (see *Vocalizations* section).

A significant time effect was found for self-grooming behavior, $F(4, 20) = 5.25$, $p < .01$, scratching behavior, $F(4, 20) = 3.28$, $p < .04$, and self-biting behavior, $F(4, 20) = 3.21$, $p < .04$. Self-directed behaviors (self-grooming, scratching, and self-biting) in both conditions increased significantly after the 1st day of separation (see Figures 2A–2C). A Condition × Time interaction effect was found for object exploration, $F(4, 20) = 3.77$, $p < .02$. A significant difference between conditions was found for object exploration on the 1st and 3rd day of testing (see Figure 2D). A Condition × Time interaction effect was found for cage-biting and cage-shaking behavior, $F(4, 20) = 4.11$, $p < .02$. Cage-biting and cage-shaking in the adjacent condition increased significantly on the last 2 days of testing. In contrast, for total separation, this behavior was at its maximum on the 1st day of separation and decreased significantly on the 2nd and last days of testing, and a significant difference between conditions was found on the last day of testing (see Figure 2E). A borderline significant Condition × Time interaction effect was found for the occurrence of hunched and freezing postures, $F(4, 20) = 2.79$, $p = .054$. A post hoc analysis showed an increase over time for the hunched and freezing postures occurring after total separation. On the last 2 days of testing, the occurrence of these postures was higher in the total separation condition compared with the adjacent one (see Figure 2F).

Vocalizations

During recording, it was apparent that the types of coo calls emitted by the infants in each of the two separation conditions were different. Two types of coos could be distinguished in the case of adjacent separation, and one type (different from the first two) could be distinguished in the case of total isolation. Because of their respective characteristics, they were named the *adjacent call modulated in frequency* (AM), the *adjacent call not modulated in frequency* (A), and the *isolation call* (T).

During the 4 days of separation, mothers in the adjacent condition were remarkably silent. Only in 2 of the 6 dyads tested did the mother even emit a few growls. Because of the rapid decline in infant calling behavior over the 4-day separation, only coo vocalizations from the first 30-min recording at the start of each separation period were subjected to spectrographic analysis for comparison.

The spectrographic analysis revealed a striking difference in the quality of coos emitted by the separated infant under the different conditions of separation. The sonagrams in Figure 3 show how the separation conditions affected the quality of calling for each infant in the study and how consistent the effects were across animals.

The spectrographic analysis of sonagrams in Figure 3 confirmed that two types of tonal calls are emitted during adjacent separation and that only one type is emitted during total isolation. These calls can be differentiated from each other on a number of acoustic measures. Table 2 gives the means and standard errors of each parameter that shows a significant difference between at least two of the three types of coos.

Parameters that differentiate the isolation coo and the adjacent coos are total duration, duration to peak frequency, duration of amplitude modulation, and total number of harmonics. The isolation coo is longer in duration and amplitude modulation than both adjacent coos. The higher number of harmonics in the isolation coo is probably linked to the higher intensity of the call in this condition. Both adjacent coos have short duration and amplitude modulation, but they differ from each other by the presence or absence of frequency modulation and the value of the fundamental at the begin-

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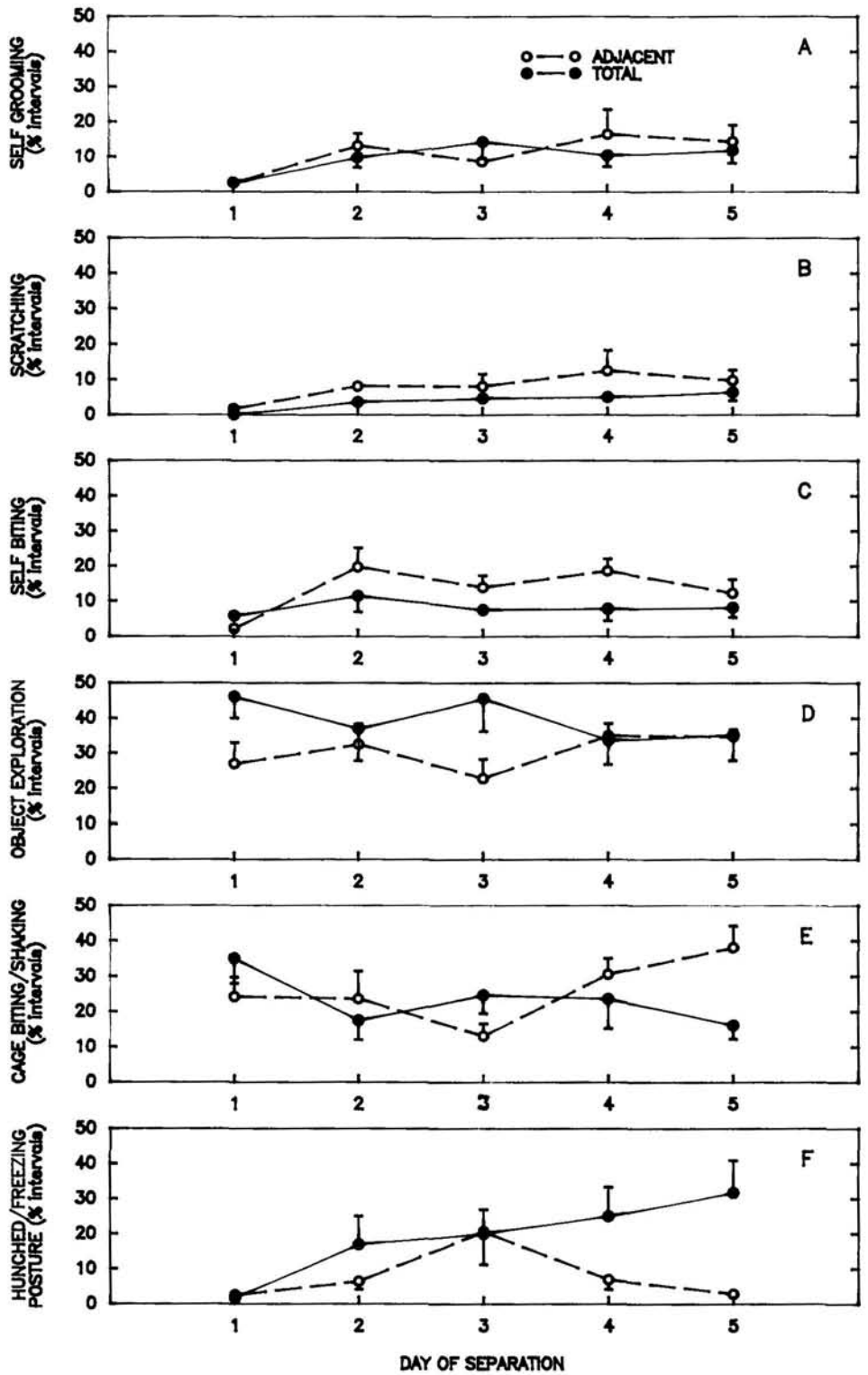
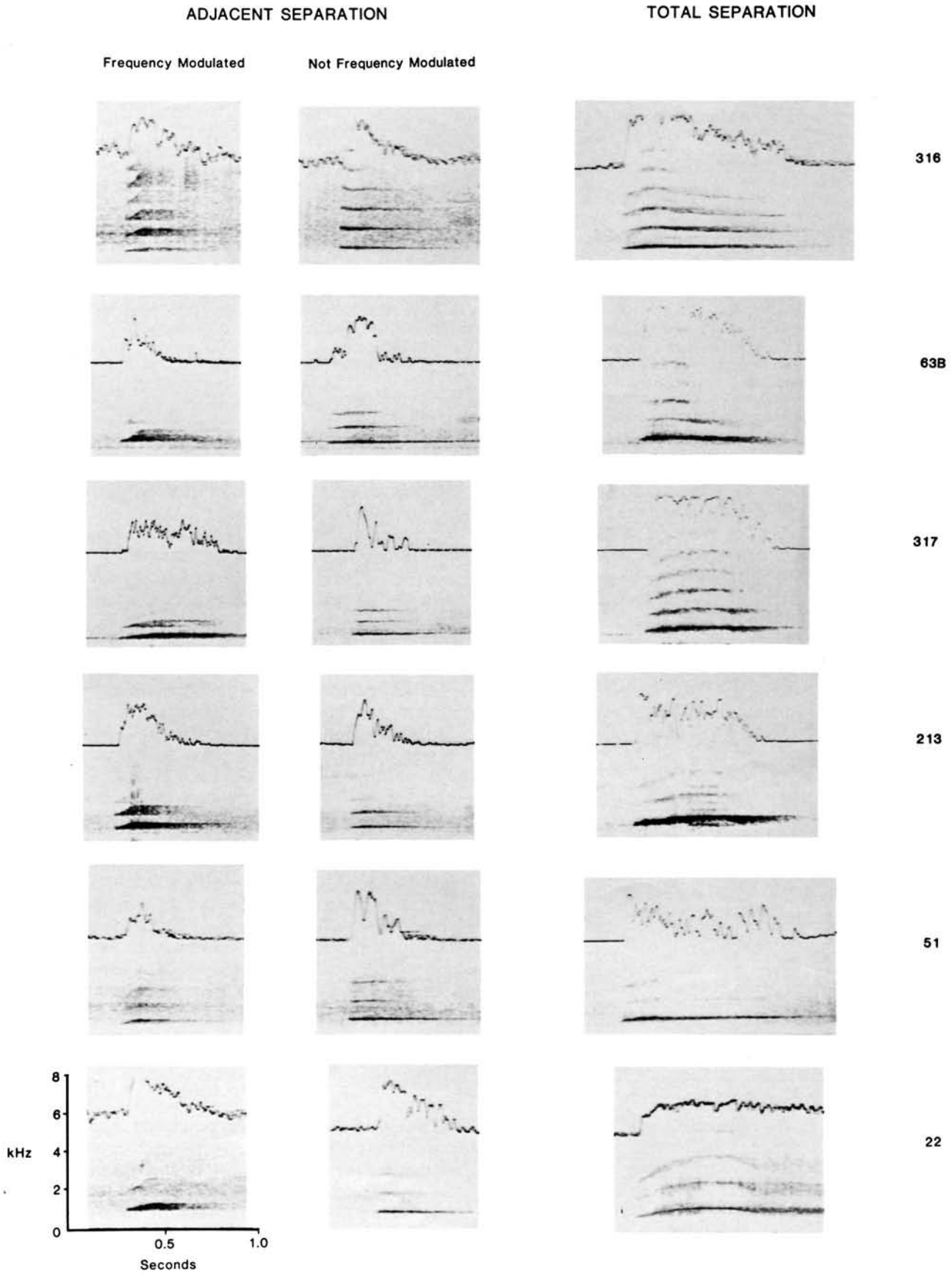


Figure 2. Infant behavioral responses to separation in adjacent and total conditions. (Data are means ± *SE*: [A] self-grooming, [B] scratch, [C] self-biting, [D] object exploration, [E] cage biting and cage shaking, [F] hunched and freezing postures.)

Figure 3. (*opposite*) Examples of coo vocalizations (whoo calls) emitted by the infants during separation in the two conditions. (The upper tracing in each panel is the amplitude modulation.)

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Table 2
Means (SE) of Acoustic Variables That Differentiate the Three Types of Coos Emitted by the Infants During Separation

Call type	Acoustic variables							
	D (ms)	DP (ms)	SF (Hz)	PF (Hz)	NH	D1 (ms)	D2 (ms)	D3 (ms)
AM								
M	474	114[a]	735[a]	976[a]	2.83	161	272	337
SE	6	3	9	15	0.09	6	9	7
A								
M	558	0[a]	838[a]	838[a]	3.16	99	204	292
SE	9	—	10	10	0.14	2	5	6
T								
M	986[b]	265[b]	811	1000[b]	4.66[b]	227[b]	440[b]	614[b]
SE	17	11	6	10	0.14	10	13	18

Note. Acoustic variables: D = duration; DP = duration to peak; SF = start frequency; PF = peak frequency; NH = number of harmonics; D1, D2, D3 = cumulative durations of the first, the first two, and the first three amplitude modulation, respectively (see Table 1 for definitions). Call type: AM = adjacent call modulated in frequency; A = adjacent call not modulated in frequency; T = (total) isolation call; $n = 72$ for each call type.
[a]Indicates significant differences between adjacent types of coos.
[b]Indicates significant differences between adjacent coos and total coos.

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ning of the call. Of the 10 acoustic measures used in the analysis, 8 differentiate between at least two of the coos, 1 (end frequency, EF) does not, and 1 (the formant) does not differentiate between categories of coos but does differentiate between some of the subjects. Table 3 gives the mean values of the formant for each individual in the study. However, this parameter should not be considered a true individual call signature, because although the analysis did show differences between subjects, there was some overlap in the formant of 4 out of the 6 infants.

Analysis of Physiological Variables

Analysis of the cortisol data, by a two-way ANOVA with repeated measures for time and condition, revealed a significant time effect, $F(2, 10) = 9.85$, $p < .01$. Plasma cortisol concentrations were elevated at 24 hr after separation when compared with base or 96-hr samples (Figure 4). There was a nearly significant interaction effect, $F(2, 10) = 3.31$, $p = .079$. In order to further clarify the trends of the data, they were additionally analyzed as differences from base. This test revealed a main effect of time, $F(1, 5) = 7.61$, $p < .04$, with the elevation above base at 24 hr greater than at 96 hr (Figure 5). There was also a main effect of condition, $F(1, 5) = 10.78$, $p < .03$, with the elevation above base in the total isolation condition greater than in the adjacent condition.

Table 3
Individual Differences in the Formant (Hz) of Coo Vocalizations

Call type	Monkey identification number					
	316	22	213	51	63	317
AM	1848	1156	990	865	967	750
A	1819	1119	977	875	750	740
T	1735	1194	1008	998	858	894
M	1801[a]	1156[a]	992[b]	913[c]	858[c]	794[b]
SE	17	11	5	21	31	25

Note. Call type: AM = adjacent call modulated in frequency; A = adjacent call not modulated in frequency; T = (total) isolation call; $n = 12$ for each call type.
[a]Indicates significant differences from all 5 other subjects.
[b]Indicates significant differences from 4 other subjects.
[c]Indicates significant differences from 3 other subjects.

Monoamines and Their Metabolites

MHPG concentrations in both separation conditions were elevated above base at 24 hr after separation, whereas at 96 hr the concentration was lower than at 24 hr but had not completely returned to basal values: main effect of time, $F(2, 10) = 6.80$, $p < .02$ (Figure 6). A main effect of condition, $F(1, 5) = 8.07$, $p < .04$, revealed that MHPG in the total isolated condition was greater than in the adjacent condition. A main effect of time was found for HVA, $F(2, 10) = 14.74$, $p < .01$, and a borderline significant main effect of time was found for 5HIAA, $F(2, 10) = 4.00$, $p = .053$. HVA concentrations were significantly elevated at 24 hr after separation when compared with base or 96-hr samples. Post hoc analysis of 5HIAA also revealed that 24-hr levels were above base and 96-hr levels. However, neither of these measures demonstrated the differences between conditions, and the response appears to be nonspecific.

Discussion

The infants showed different behavioral profiles during the two separation conditions. The presence or absence of the mother affected not only the vocalization rates of the infants but also the amount of object exploration, cage-biting and cage-shaking behavior, and the occurrence of hunched and freezing postures. We found that immediately after separation, when the mother was absent, the infants vocalized less but showed more explorative and agitated behavior (cage

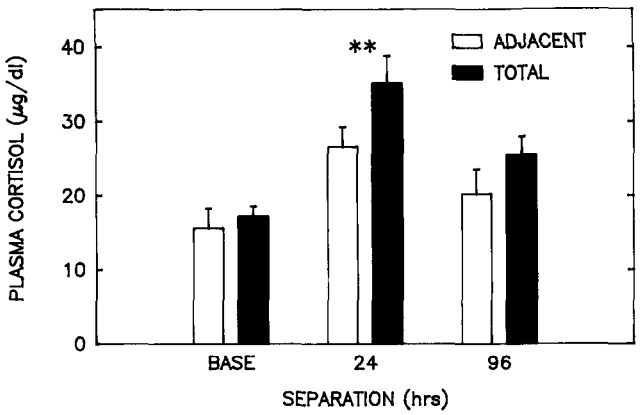


Figure 4. Infant plasma cortisol levels (means + *SE*) before and after separation in the two conditions (**24-hr samples > base = 96 hr).

biting and cage shaking). Toward the end of the 4-day separation period, the infants in proximity to their mothers became agitated, whereas the totally isolated infants became more depressed, spending more time in the hunched and freezing postures. Also of interest is the dramatic change in behavior that occurred in both conditions on the 2nd day of separation. After 24 hr separation, calling behavior (for adjacent) and agitated behavior (for total isolation) dropped significantly, whereas self-directed behaviors started to increase in both conditions. Although not explicitly recorded, infant attention during calling seemed to vary according to the presence or absence of the mother. The totally isolated infants were observed to be visually scanning their environment during and between calls, perhaps in expectation of some social feedback. In contrast, the adjacent separated infants rarely looked around when they called, presumably because of the presence of the mother. Perhaps one of the most striking behavioral difference between the two separation conditions was the quality of the infants' coo vocalizations in relation to the presence or absence of the mother.

Although rhesus macaques have been the traditional subject for studies on the effects of maternal separation (Mineka & Suomi, 1978), and although infant calling has been reported to be the major change in behavior shortly after separation, no studies on rhesus infant vocalization, per se, have been done under such circumstances. In fact, little is known about the vocal repertoire of rhesus infants; since Rowell and Hinde (1962) first described the calls of infant rhesus macaques, only a study on vocal pathology in socially deprived monkeys by Newman and Symmes (1974) has been reported. However, clear individual differences as well as two classes of infant coos have been found in young stump-tail macaques: one given when they were alone and one given as they approached their mothers within a free-ranging group (Lillehei & Snowdon, 1978). It is also worthwhile to note that the description in the literature of the primate infant "isolation" call has been very confusing:

> The detailed nature of a species' isolation call is often unclear from behavioral accounts. This is due, in part, to diverse terminology. Infants separated from their mothers are variously said to utter *distress cries, lost calls, long calls,* or *isolation calls.* Perhaps best of all would be *separation calls,* because these encompass the situation of only partial isolation. However, *isolation call* best describes the context in which these sounds occur. (Newman, 1985, p. 312)

This study clearly demonstrates that the presence of the mother affects not only the quantity but also the quality of the vocalizations emitted by infant rhesus monkeys during separation. We found that two types of coo vocalizations are emitted by the infants during adjacent separation from the mother and that a third totally different type of coo is emitted by the infants during total separation. Moreover, we found that the production of these different types of coo calls, exclusive to the separation conditions (adjacent vs. total), corresponds to different levels of arousal on the part of the infants when separated in the two conditions (see hormonal and brain monoamine data). Because of the context in which these calls occur, we found that the term *isolation call* best describes the coos emitted during total separation and that *adjacent separation call* best describes the coos emitted during the adjacent separation condition. This finding is discussed both in terms of the structural characteristics of these calls and in terms of their possible communicative function.

The fact that coo vocalizations discriminate between separation conditions is not surprising. Coos have already been described for a large array of macaque species. They are usually emitted by both sexes at all ages in different social contexts and are best known as calls that promote social contact between the different members of a troop. There is evidence in the literature that different social cues can elicit different types of coos. For example, Green (1975) reported seven classes of coos that are associated with different social contexts in the Japanese macaque.

There are several reasons why coos may be important in the discrimination between mother present and mother absent during separation. First, coos are tonal in structure compared with other calls that are harsh or nontonal, and it has been reported that tonal calls, especially the ones modulated

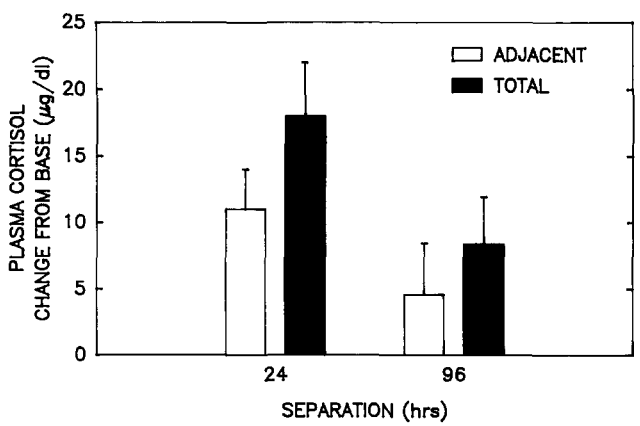


Figure 5. Mean (+ *SE*) differences from base values of cortisol 24 and 96 hr after separation in the total isolation and adjacent conditions (24 hr > 96 hr; total isolation > adjacent).

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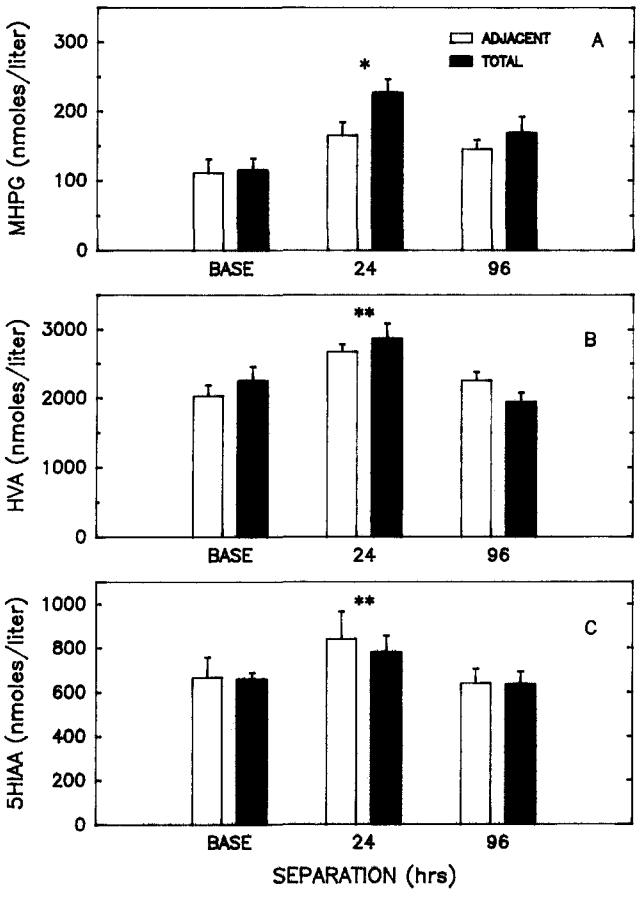


Figure 6. Mean (+ *SE*) concentrations of monoamine metabolites in infants' cerebrospinal fluid before and after separation in the two conditions. ([A] MHPG [norepinephrine metabolite; total isolation > adjacent, *24 hr > base], [B] HVA [dopamine metabolite; **24 hr > base = 96 hr], [C] 5HIAA [serotonin metabolite; **24 hr > base = 96 hr].)

in frequency, may facilitate localization of the sender (Brown, 1982; Snowdon & Hodun, 1981). Second, the coos emitted in the two separation conditions have major differences in acoustical properties that may be used for discriminative purposes. The isolation call of long duration, high intensity, and high frequency modulation may be viewed as a long-distance call. In contrast, the adjacent separation calls that are shorter in duration, lower in intensity, and have short amplitude modulation are best representative of short-distance calls. The general assumption that the length and amplitude of a call increase according to the level of arousal of the sender is partially confirmed by this experiment, which shows a comparatively higher level of arousal in the infants that produce the isolation call. However, because these calls have strictly different configurations, they may have different symbolic communicative value as well.

Several hypotheses can be raised regarding the function of these calls. Because vocal behavior is one of the most important means of communication in primates, different calls given in various separation conditions must play an important role in the infant's life, especially within the context of the mother–infant relationship. It is probably of great survival value for the infant to have an isolation call that is acoustically different from a separation call. These types of calls may convey different information to the mother, which in turn may react differently toward the infant. Two assumptions can be made about the respective roles played by the isolation call and the adjacent separation calls in a more natural environment. In the case in which mother and infant are not in view of each other, the isolation call may indicate that the infant is lost and may facilitate localization and retrieval of the infant by the mother. In the case in which the infant is beyond reach of the mother, but in visual contact with her, the adjacent separation calls may indicate a sudden need for contact and reassurance, upon which the mother then has the choice of retrieving the infant or not. Thus, the isolation call (a long-distance call) may be viewed as an indicator of an emergency situation, in which it functions to ensure the survival of the infant, whereas the adjacent separation calls (short-distance calls) may serve the particular function of metacommunication between mother and infant.

Whether these calls might have meaning for group members other than the mother is not known. However, given the fact that for the different primate species tested, mothers are able to recognize the vocalizations of their own offspring, especially in the case of isolation calls (Cheney & Seyfarth, 1980; Symmes & Biben, 1985) or juvenile coos (Pereira, 1986), one can hypothesize that natural selection has favored the production of these calls to be mostly effective in the context of the mother–infant relationship. It is also very likely that the mother, other group members, or both can discriminate between the two types of calls described in this study, because there is now some evidence that macaques are able to perceive distinctively two variants of coos that are acoustically different and usually emitted in different social contexts (Petersen, 1982).

The cortisol results are consistent with what has been previously reported (Levine et al., 1985). Totally isolated infants tend to show higher and more protracted plasma cortisol elevations than infants separated in the adjacent condition. The inverse relation between vocalization and cortisol is predicted by the coping model that would anticipate that the vocalization would attenuate the cortisol (stress) response in infants (Levine, 1983).

Interestingly, the monoamine metabolites show a similar general response (24-hr elevation) to separation as cortisol, with MHPG (the metabolite of norepinephrine) also discriminating between the adjacent and total separations. Although more experimentation is needed to clarify the relationship, if any, between pituitary–adrenal activity and brain norepinephrine activity, we can conclude from this experiment that both systems are activated during maternal separation and that, for both systems, the response is specific to the separation condition.

Both cortisol and norepinephrine activity reflect the different levels of stress experienced by the infants under the two separation conditions. Different behavioral responses may be responsible for the different stress levels under the two separation conditions. Theoretically, under both conditions the infant has the ability, through calling, to establish

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reunion with the mother. During total isolation, the nature of the infant's vocalizations clearly indicates that complete absence of contact leads to a highly stressful response. In this context, the lack of social feedback may be responsible for the rapid cessation of calling and agitated behavior observed shortly after separation. Furthermore, the amount of stress expressed physiologically by the infant as a consequence of the total lack of social feedback may later result in the development of a depressive state. When the mother is in proximity but out of reach, the infant spends more time calling, and the nature of the calls is different and appears to be directly related to the presence of the mother. In this context, both the quantity and the quality of vocalizations may be important in the process of reducing stress when there is some social feedback during separation. We have shown that the mother provides appropriate social feedback in this experiment; however, it remains to be tested whether a conspecific would be satisfactory.

This study clearly demonstrates that the infant's response to separation is a complex phenomenon subject to change with different social or environmental cues, or both; in particular, the presence of the mother during separation affects the infant's cognitive and affective responses to separation at both the behavioral and neuroendocrinological levels.

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